

Scalability & Future Plan

Date	2 July, 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (H/M/L)
1	Single-model architecture on local server	Limited scalability for global banking operations	High
2	Lack of real-time explainability logic	Users receive decisions without understanding the driving factors (income, history)	Medium
3	Manual notification for borderline cases	Increased workload for compliance officers and slower turnaround times	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	Model Deployment	Python/Scikit-learn local instance	Deploy on IBM Watson Machine Learning for high-availability and API-based scaling
2	Feature Processing	Synchronous data transformation	Implement Spark-based feature engineering for handling massive applicant record batches

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
3	User Traffic	Flask development server	Migrate to Gunicorn/Nginx on IBM Cloud with auto-scaling for concurrent predictions
4	Dashboard/UI	Static Matplotlib charts	Integrate interactive Plotly/D3.js dashboards for real-time risk visualization

Future Roadmap:

Phase	Planned Feature / Enhancement	Target timeline	Expected Impact
Phase-2	Model Explainability (SHAP/LIME)	04/07/2026	Improved transparency for compliance and customer trust
Phase-3	Interactive Sandbox Mode	05/07/2026	Allows customers to test 'what-if' scenarios for credit health improvement
Phase-4	Automated Compliance Alerts	06/07/2026	Real-time notifications to analysts for manual review of high-value cases